

Foundations of Artificial Intelligence

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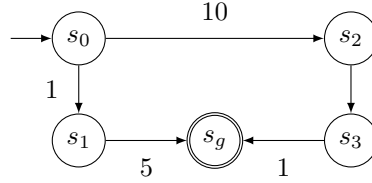
Exercise Sheet 5 — Solutions

Exercise 5.1 (1+1 marks)

Execute

- (a) greedy best-first search ($f(n) = h(n.state)$) and
- (b) A* ($f(n) = g(n) + h(n.state)$)

without reopening in the state space depicted below. As heuristic, use the perfect heuristic h^* .



Describe the execution of both search algorithms with the following schema:

1. expanding s_{10} : $open = \langle s_{11}(f = 5), s_{12}(f = 6) \rangle$, $closed = \{s_{10}(g = 0)\}$
2. expanding s_{11} : $open = \langle s_{13}(f = 4), s_{12}(f = 6), s_{16}(f = 10) \rangle$, $closed = \{s_{10}(g = 0), s_{11}(g = 3)\}$
- ...
- i. expanding s_i : found goal with cost x

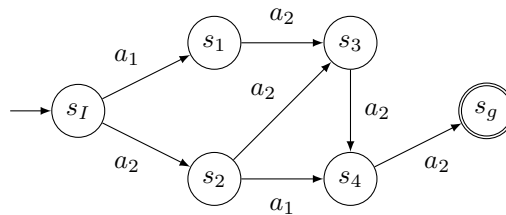
Note: This schema uses fictitious states and numbers that don't correspond to the given state space.

Solution:

- (a)
 1. expanding s_0 : $open = \langle s_2(f = 2), s_1(f = 5) \rangle$, $closed = \{s_0(g = 0)\}$
 2. expanding s_2 : $open = \langle s_3(f = 1), s_1(f = 5) \rangle$, $closed = \{s_0(g = 0), s_2(g = 10)\}$
 3. expanding s_3 : $open = \langle s_g(f = 0), s_1(f = 5) \rangle$, $closed = \{s_0(g = 0), s_2(g = 10), s_3(g = 11)\}$
 4. expanding s_g : found goal with cost 12
- (b)
 1. expanding s_0 : $open = \langle s_1(f = 6), s_2(f = 12) \rangle$, $closed = \{s_0(g = 0)\}$
 2. expanding s_1 : $open = \langle s_g(f = 6), s_2(f = 12) \rangle$, $closed = \{s_0(g = 0), s_1(g = 1)\}$
 3. expanding s_g : found goal with cost 6

Exercise 5.2 (2 marks)

Consider the following state space $\mathcal{S}$ where $cost(a_1) = cost(a_2) = 2$:



Define an admissible heuristic h for $\mathcal{S}$ such that IDA* behaves in the following way:

- IDA* performs at least 4 iterations and
- exactly one new path is reachable within the new f limit in each iteration (simulating worst-case behavior of IDA*).

Provide your heuristic h and for each iteration the f limit as well as the newly explored path.

Hint: There are solutions which use only natural numbers (including 0) as heuristic values. To satisfy the second requirement, expanded nodes must have distinct f -values.

Solution:

$$h = \{s_I \mapsto 0, s_1 \mapsto 1, s_2 \mapsto 2, s_3 \mapsto 3, s_4 \mapsto 1, s_g \mapsto 0\}$$

iteration	f limit	new path
1	0	$\langle \rangle$
2	3	$\langle a_1 \rangle$
3	4	$\langle a_2 \rangle$
4	5	$\langle a_2, a_1 \rangle$
5	6	$\langle a_2, a_1, a_2 \rangle$

Exercise 5.3 (2 marks)

In Chapter B14 we showed that A^* with reopening is optimal when using admissible heuristics. If you would try to apply the proof to weighted A^* instead of A^* , where would it fail? Justify why it fails at the point you specify and briefly explain why it does not fail earlier.

Solution:

It would fail at the f -Bound Lemma. The optimal continuation lemma still holds with weighted A^* ; in fact it holds with any best-first search algorithm that only prunes states s with $h(s) = \infty$ as long as we use a safe heuristic: In such a setting iteration B is guaranteed to insert the required node into open and as long as s' is not settled this node will stay in open.

The f -Bound Lemma on the other hand does not hold anymore. More specifically, the inductive proof does not hold for weighted A^* : Even for the base case we have that $f(n_0) = w \cdot h(s_0)$ can be larger than c^* , for example it is violated with $w > 1$ and $h = h^*$. Similarly in the induction step we have that $f(n') = g^*(s_i) + w \cdot h(s_i)$ might be larger than $g^*(s_i) + h^*(s_i)$.

Exercise 5.4 (2+2 marks)

Show with a counterexample that the following two statements are not correct:

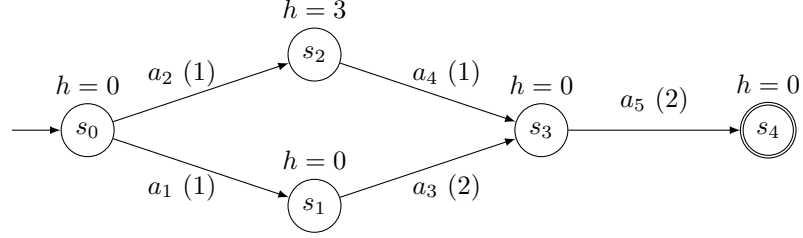
- A^* without reopening using an admissible but inconsistent heuristic is optimal.
- A^* without reopening using a consistent but inadmissible heuristic is optimal.

For each subtask, the counterexample must consist of a state space and a heuristic. Provide the state space as a directed graph, similar to the “bounded inc-and-square” state space introduced in Chapter B1. Add the action costs and the heuristic values as additional labels to the edges and nodes, respectively, you don’t need to write them down separately. Show that the heuristic has the required properties, and show that A^* is not optimal. The solution found should be suboptimal independently of tie-breaking, that is, it should not depend on which states with the same f -value are expanded first.

Hint: For one of the tasks (a) or (b), the counterexample requires multiple paths to the same state. For the other, the counterexample requires more than one goal state.

Solution:

(a) Consider the following state space:

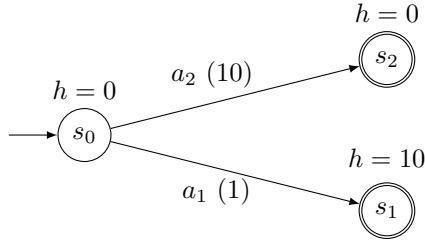


The heuristic is clearly admissible (the only non-zero value is $h(s_2) = 3 = h^*(s_2)$), but not consistent ($h(s_2) \not\leq h(s_3) + \text{cost}(a_4)$). A* without reopening will find a solution with cost 5, while the optimal solution $\pi = \langle a_2, a_4, a_5 \rangle$ has cost 4.

The following table depicts the end of each iteration of the main while loop. A node is represented as a tuple containing the state and f -value. Note that in step 5 no expansion is performed, since we consider A* *without* reopening and we already expanded a node containing s_3 .

step	state	open list	closed list
1	s_0	$\langle \langle s_1, 1 \rangle, \langle s_2, 4 \rangle \rangle$	$\{s_0\}$
2	s_1	$\langle \langle s_3, 3 \rangle, \langle s_2, 4 \rangle \rangle$	$\{s_0, s_1\}$
3	s_3	$\langle \langle s_2, 4 \rangle, \langle s_4, 5 \rangle \rangle$	$\{s_0, s_1, s_3\}$
4	s_2	$\langle \langle s_3, 2 \rangle, \langle s_4, 5 \rangle \rangle$	$\{s_0, s_1, s_2, s_3\}$
5	s_3	$\langle \langle s_4, 5 \rangle \rangle$	$\{s_0, s_1, s_2, s_3\}$
6	s_4	solution: $\langle a_1, a_3, a_5 \rangle$ (cost: 5)	

(b) Consider the following state space:



The heuristic is consistent (since $h(s_0) \leq h(s_1) + 1$ and $h(s_0) \leq h(s_2) + 10$) holds) but inadmissible ($h^*(s_1) = 0$, since s_1 is a goal state, but $h(s_1) = 10$).

A* (both with and without reopening) will first expand $\langle s_0, 0 \rangle$, inserting $\langle s_1, 11 \rangle$ and $\langle s_2, 10 \rangle$ into the open list. In the next step, $\langle s_2, 10 \rangle$ is expanded, and since s_2 is a goal state, the algorithm returns $\langle a_2 \rangle$ with cost 10, but the optimal solution is $\langle a_1 \rangle$ with cost 1.